

# Quadratic equations using technology



1

a  $m = 0, -5$

c  $v = \frac{8}{5}, -\frac{8}{5}$

e  $k = \frac{4}{9}, -\frac{4}{9}$

g  $y = 5, -5$

i  $k = 2, -2$

b  $m = 0, -5$

d  $x = 8, -8$

f  $p = -10, 10$

h  $y = \frac{6}{5}, -\frac{6}{5}$

2

a  $x = \frac{7}{8}, -\frac{25}{8}$

c  $x = 0, -7$

e  $x = -\frac{8}{3}, \frac{7}{5}$

g  $x = \pm 7\sqrt{6}$

i  $x = \frac{9}{10}$

k  $x = 0, \frac{9}{2}$

m  $x = \pm\sqrt{2}$

o  $x = 4, -10$

q  $x = 6 \pm \sqrt{2}$

s  $x = \pm 11$

u  $x = \pm 5$

b  $x = 4, 2$

d  $x = \frac{5}{8}, \frac{7}{3}$

f  $x = \frac{17}{3}, -\frac{d}{2}$

h  $x = \frac{4}{9}, \frac{-4}{9}$

j  $x = \frac{3}{7}$

l  $x = 6, -7$

n  $x = 11, -5$

p  $x = -7, 11$

r  $x = 5, -5$

t  $x = \pm 11$

3

a  $x = 2, x = -\frac{5}{3}$

b  $x = \frac{3}{2}, -\frac{7}{2}$

c  $x = 0.4, -2.0$

4

a  $x = \pm\sqrt{2}$

b  $x = 1.4, -1.4$

5

a 1

b 1

6

$x = -\frac{3}{5}, -2, -\frac{5}{2}, -4$

## Applications



7  $t = 14, t = 10$

8

a  $x = 0, 32$

b 32 cm

9

a  $17 + \frac{6426}{x^2} = \frac{663}{x}$

b  $x = 18, 21$

c 36.83 cm